

be calculated as per Eq. 1.

More elaborate definitions of electric and magnetic field strength/intensity are given in the Box.

The concept of power density at distance D from an isotropic radiator radiating power P in space will be clear from the spherical model of free space with an isotropic antenna at its centre as shown in Fig. 9.

If one uses any other type of antenna with known gain, the power density in the line of shoot (maximum power) can be obtained by multiplying the power density due to the isotropic radiator by true antenna gain for the specific antenna (refer Fig. 10) used.

Conversion between units and sub-units. Table I shows conversions between power density units expressed as W/m², mW/cm² and μW/cm².

Practical base station antennae (Fig. 11). Several types of antennae are used for cellphone communications; e.g., panel-shaped sector antennae or pole-shaped omnidirectional antennae. Dish antennae form terminals for point-to-point microwave links that communicate with other base stations and link the network together. Sometimes the base stations are connected together using buried cables instead of microwave links.

Beam shapes and directions. The power from antennae used with macrocellular base stations is radiated in conical fan-shaped beams directed towards the horizon with a slight downward tilt (see Fig. 12). The EM wave strengths below the antennae and at the base of masts are much lower than directly in front of the antennae.

The beams from the antennae spread out with distance and tend to reach ground level at distances in the range of 50-300 metres from the antennae. The power levels at distance D can be calculated easily using Eq. (4) in Fig. 9 if power fed into the antenna and antenna gain are known.

Typically, base station towers or structures are 15-50m high. A cellphone base station may utilise several

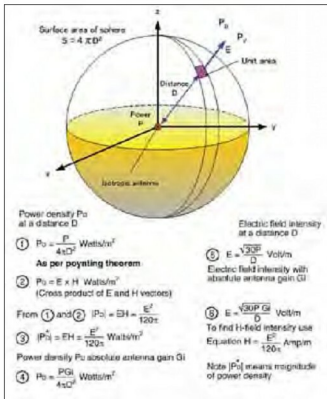


Fig. 9: EM radiation from isotropic antenna and its equations

Type of antenna	Absolute gain	
	True value (times) G ₀	Decibels value
Isotropic	1	0
Hertzian Dipole	1.5	1.76
$\lambda/2$ Dipole	1.64	2.15
$\lambda/4$ Monopole	3.28	5.15
GSM 65 degree Sectorial	50.12	17

Fig. 10: Absolute gain true and dB values of typical antennae



Fig. 11: Practical base station antennae

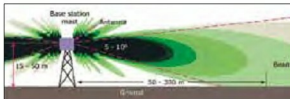


Fig. 12: Typical cellular antenna beam shape

omnidirectional antennae. In urban areas, cellphone service providers commonly use sector antennae for their

base stations. These antennae are rectangular panels, about 30×125 cm in dimension.

The total RF power that could be transmitted from each transmitting antenna at a cell site depends on the number of radio channels (transmitters) that have been authorised and the power of each transmitter. Typically, for a cellphone base station, a maximum of 21 channels per sector (depending on the system) could be used. Thus for a typical cell site utilising sector antennae, each of the three transmitting antennae could be connected to

up to 21 transmitters (or a total of 63 transmitters per site). Furthermore, while a typical base station could have as many as 63 transmitters, not all of the transmitters would be expected to operate simultaneously, thus reducing overall emission levels. The maximum power radiated in any direction usually does not exceed 50 watts.

Macro, micro and pico cells. Base stations communicate with mobile phones within a defined area or 'cell.' There can be three types of cells—macro, micro and pico cells—depending upon their size and the radiated power from the antenna.

Macro cells provide the main structure for the base station network with power outputs of tens of watts to communicate with cell phones up to about 35 km. These would have higher level of EM radiation and hence require more stringent monitoring.

Micro cells are used to infill and improve the main network, especially where the volume of traffic (calls) is high. These are good for